

Suggested FYP topics for 2026-27

by

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Project code :	AG1
Project title :	An Information-Theoretic Inequality for the Information Bottleneck
No. of students :	2
Description of project :	Recently, an information-theoretic inequality involving three Boolean random variables was conjectured. This conjecture is closely connected to the information bottleneck principle in machine learning and extends Ms. Gerber's Lemma from classical information theory. In this project, students will explore a range of optimization-based approaches to investigate the conjecture. They are also encouraged to develop potential generalizations and related extensions of the result.
Industrial collaboration :	No.

Project code :	AG2
Project title :	f-divergence information inequalities
No. of students :	2
Description of project :	Shannon's mutual information is a fundamental concept in information theory, and is utilized to quantify the dependency and shared information between two random variables. This measure serves as a cornerstone for various applications, including communication systems, machine learning, and statistical inference. The mutual information can be generalized through the framework of f-divergence, which provides a broader perspective on measuring the divergence between probability distributions. The primary objective of this project is to conduct numerical simulations aimed at identifying a comprehensive list of feasible information inequalities that are satisfied by information measures defined via f-divergence. By exploring these inequalities, the project seeks to deepen our understanding of the relationships and constraints between different information-theoretic measures. While the project is primarily simulation-based, students will also engage with the foundational concepts of information theory. They will learn about key topics such as entropy, Kullback-Leibler divergence, and the properties of f-divergences.
Industrial collaboration :	No.

Project code :	AG3
Project title :	Self-proposed topic

No. of students :	2
Description of project :	-
Industrial collaboration :	No.